1. (习题 6-25)

$1 \to 2$, 考虑可逆等温膨胀过程,

$$\Delta S_{1\to 2} = \int_1^2 \frac{\delta Q}{T} = \frac{1}{T_1}\int_1^2 \mathrm{d}A = \frac{1}{T_1}\int_1^2 \frac{nRT_1\mathrm{d}V}{V} = nR\ln\frac{V_2}{V_1},$$
$$\Delta S_{1\to 2} \approx 5.76\mathrm{J/K}.$$

$1 \to 3 \to 2$, 氢气有 $i = 5$; 分别考虑可逆等压膨胀和可逆等体膨胀,

$$\begin{aligned}\Delta S_{1\to 3} &= \int_1^3 \frac{\delta Q}{T} \\ &= \int_1^3 \frac{\mathrm{d}E}{T} + \int_1^3 \frac{p\mathrm{d}V}{T} \\ &= \int_1^3 \frac{\frac{5}{2}nR\mathrm{d}V}{V} + \int_1^3 \frac{nR\mathrm{d}V}{V} \\ &= \frac{7}{2}nR\ln\frac{V_2}{V_1};\end{aligned}$$

此时温度满足 $\frac{T_3}{T_1} = \frac{V_3}{V_1} = \frac{V_2}{V_1}$.

$$\begin{aligned}\Delta S_{3\to 2} &= \int_3^2 \frac{\mathrm{d}E}{T} \\ &= \int_3^2 \frac{\frac{5}{2}nR\mathrm{d}T}{T} \\ &= \frac{5}{2}nR\ln\frac{T_2}{T_3} \\ &= -\frac{5}{2}nR\ln\frac{V_2}{V_1}.\end{aligned}$$

因此 $\Delta S_{1\to 3\to 2} = nR\ln\frac{V_2}{V_1} = \Delta S_{1\to 2}$.

$1 \to 4 \to 2$, 氢气有 $\gamma = \frac{7}{5}$; 前者绝热无熵变, 过程前后满足 $p_1V_1^\gamma = p_2V_4^\gamma$, 因此

$$\ln\frac{V_4}{V_1} = \frac{1}{\gamma}\ln\frac{p_1}{p_2} = \frac{1}{\gamma}\ln\frac{V_1}{V_2} \Rightarrow \ln\frac{V_2}{V_4} = -\ln\frac{V_4}{V_2} = -\left(\ln\frac{V_4}{V_1} + \ln 2\right) = \frac{2}{7}\ln 2.$$

后者仍考虑可逆等压膨胀, 形如上述 $\Delta S_{1\to 3}$ 可知

$$\Delta S_{4\to 2} = nR\ln\frac{V_2}{V_1} = \Delta S_{1\to 2}.$$

结论: S 是状态量, 与平衡态间转移的过程无关.

2. (习题 6-27)

已知常数有

$$C_{\mathrm{water},p} = 4.22\times 10^3\mathrm{J/(kg\cdot K)}, \quad C_{\mathrm{ice},p} = 2.09\times 10^3\mathrm{J/(kg\cdot K)},$$
$$\lambda_{\mathrm{ice}} = 3.34\times 10^5\mathrm{J/kg}.$$

(1) (若只计算水的熵变:)

$$
\begin{aligned}
\Delta S=\int_1^3 \frac{\delta Q}{T} &= \int_{-10^{\circ}\mathrm{C}}^{0^{\circ}\mathrm{C}} \frac{mC_{\mathrm{ice},p}\mathrm{d}T}{T}+\frac{m\lambda_{\mathrm{ice}}}{T_0}+\int_{0^{\circ}\mathrm{C}}^{10^{\circ}\mathrm{C}} \frac{mC_{\mathrm{water},p}\mathrm{d}T}{T} \\
&= mC_{\mathrm{ice},p}\ln\frac{273}{263}+\frac{m\lambda_{\mathrm{ice}}}{273\mathrm{K}}+mC_{\mathrm{water},p}\ln\frac{283}{273} \\
&\approx 1453\mathrm{J/K}.
\end{aligned}
$$

(若假设环境恒温 $10^{\circ}\mathrm{C}$, 则还需要加上环境熵变:)

$$
\begin{aligned}
\Delta S_{\mathrm{env}}=\int_1^3 \frac{\delta Q}{T} &= \frac{-mC_{\mathrm{ice,p}}\Delta T}{T_{\mathrm{env}}}+\frac{-m\lambda_{\mathrm{ice}}}{T_{\mathrm{env}}}+\frac{-mC_{\mathrm{water},p}\Delta T}{T_{\mathrm{env}}} \\
&= -1403\mathrm{J/K},
\end{aligned}
$$

$$
\Delta S_{\mathrm{total}}=\Delta S+\Delta S_{\mathrm{env}}\approx 50\mathrm{J/K}.
$$

(2) 冰变为水前后, 水的熵变为

$$
\Delta S_0=\frac{m\lambda_{\mathrm{ice}}}{273\mathrm{K}}\approx 1223\mathrm{J/K}.
$$

根据 Boltzmann 熵的定义,

$$
\ln\frac{W_{\mathrm{water}}}{W_{\mathrm{ice}}}=\frac{\Delta S_0}{k}.
$$

可知

$$
\frac{W_{\mathrm{water}}}{W_{\mathrm{ice}}}=\exp(8.86\times 10^{25}).
$$

3. (熵的概念理解)

熵是系统微观状态数的度量, 熵越大, 系统所对应的微观状态数越多 (常说的 "更无序"). 它也在宏观层面描述了热量交换与温度的关系.

4. (补充题)

气体依次经历

$$
(p_0,V_0,T_0)\to(2p_0,V_0,2T_0)\to(p_0,2V_0,2T_0).
$$

不考虑容器体积对自由程的约束, 可知

$$
\frac{\overline{\lambda}}{\overline{\lambda_0}}=\frac{Tp_0}{T_0p}=2.
$$

$$
\kappa=\frac{1}{3}m_0n\cdot\overline{v}\cdot\overline{\lambda}\frac{C_{V,m}}{M}\propto\overline{v}\propto\sqrt{T}\Rightarrow\frac{\kappa}{\kappa_0}=\sqrt{2}.
$$

同理,

$$
\frac{\eta}{\eta_0}=\sqrt{2}.
$$

$$
D=\frac{1}{3}\overline{v}\overline{\lambda}\propto\frac{T^{\frac{3}{2}}}{p}\Rightarrow\frac{D}{D_0}=2\sqrt{2}.
$$

$$
z
$$